Statistical Modelling & Analysis (STAT2004/2904/7004)
Assignment 1

Submit your solutions via Blackboard as a single PDF file.

Due Friday 21 August at 4:00 pm

STAT2004/7004: Complete Exercises 1–4 for a maximum of 40 marks and a total of 10%.
STAT2904: Complete Exercises 1–5 for a maximum of 50 marks and a total of 10%.

1. (10 marks) The Rayleigh distribution is often used to model nonnegative magnitudes, such as wind speeds, wave heights, or signal amplitudes, when the underlying quantity can be viewed as the magnitude of a two-dimensional random vector. The probability density function for this distribution is

$$f(x;\theta) = \frac{x}{\theta}\exp\left(-\frac{x^2}{2\theta}\right), \qquad x > 0,\ \theta > 0.$$

Let $X_1, \ldots, X_n$ be a random sample from this distribution. For parts (d) and (e) it may help to know that

$$\mathbb{E}_\theta(X^2) = 2\theta, \qquad \mathbb{E}_\theta(X^4) = 8\theta^2.$$

(a) Write down the log-likelihood function for this model. [1 mark]

(b) Write down the score function, that is the derivative of the log-likelihood function with respect to θ. [1 mark]

(c) Determine the maximum likelihood estimator of θ. Make sure you establish that you have the global maximum of the likelihood. [2 marks]

(d) Determine the bias of the maximum likelihood estimator of θ. [2 marks]

(e) Determine the mean squared error of the maximum likelihood estimator of θ. [3 marks]

(f) Is the maximum likelihood estimator of θ a consistent estimator? Justify your answer. [1 mark]

2. (10 marks) The zero-truncated Poisson distribution is used to model count data where only positive counts can be observed, or where zero counts are excluded from the sample. The probability mass function for this distribution is

$$f(x;\theta) = \frac{\theta^x}{x!\{\exp(\theta) - 1\}}, \qquad x \in \{1, 2, \ldots\},\ \theta > 0.$$

Let $X_1, \ldots, X_n$ be a random sample from the zero-truncated Poisson distribution.

(a) Write down the log-likelihood function for this model. [1 mark]

(b) Write down the score function and show that the maximum likelihood estimator must satisfy the equation $g(\theta) = \bar{X}$, where $g : (0, \infty) \to (0, \infty)$ is the function defined by

$$g(\theta) = \frac{\theta}{1 - \exp(-\theta)}$$

and $\bar{X} = n^{-1}\sum_{j=1}^{n} X_j$. [2 marks]

(c) Show that the function g is strictly increasing on $(0, \infty)$. [3 marks]

(d) When does the maximum likelihood estimator exist? [3 marks]

(e) Assuming the maximum likelihood estimator exists, is it unique? Justify your answer. [1 mark]

3. (10 marks)The Lomax distribution is a heavy-tailed distribution often used to model positive-valued data such as insurance losses, lifetimes, waiting times, or other skewed data with occasional large observations. The probability density function for this distribution is

$$f(x;\theta) = \frac{\alpha}{\lambda}\left(1+\frac{x}{\lambda}\right)^{-(\alpha+1)}, \qquad x > 0, \quad \theta = (\alpha, \lambda) \in (0,\infty)^2.$$

The mean and variance of the Lomax distribution are

$$\mathbb{E}_\theta(X) = \begin{cases} \dfrac{\lambda}{\alpha - 1}, & \alpha > 1, \\ \infty, & 0 < \alpha \leq 1, \end{cases}$$

and

$$\mathrm{Var}_\theta(X) = \begin{cases} \dfrac{\lambda^2 \alpha}{(\alpha-1)^2(\alpha-2)}, & \alpha > 2, \\ \infty, & 0 < \alpha \leq 2. \end{cases}$$

Let $X_1, \ldots, X_n$ be a random sample from this distribution.

(a) Identify one difficulty with applying the method of moments for this distribution. [1 mark]

(b) Assuming the relevant moments exist and are finite, provide expressions for the method of moments estimator of

$$\theta = (\alpha, \lambda).$$

Write your answer in terms of the sample mean and sample variance. You need to identify any condition(s) required for the method of moments estimator to give a value in the parameter space $(0, \infty)^2$. [4 marks]

(c) Write down the log-likelihood function for this model. [1 mark]

(d) The file `lomax.csv` provides 1000 observations. In R or Python, write a suitable function to evaluate the log-likelihood for this model. Use `optim` in R (or an equivalent numerical optimiser in Python) to find the maximum likelihood estimate for this data. You may want to use the method of moments estimator from part (b) as an initial value.
You must show your code and the output from `optim` (or equivalent optimiser in Python).
[4 marks]

Note: You can read the data into R using the following code:

```
data <- scan("lomax.csv")
```

assuming that the file `lomax.csv` is in the working directory.

4. (10 marks) Let $X_1, \ldots, X_n$ be a random sample from a $\mathsf{Binomial}(2, p)$ distribution, where $p \in [0, 1]$. We are interested in estimating p^2, the probability of two successes from two trials.

   (a) Define random variables $Y_1, \ldots, Y_n$ by

   $$Y_j = \begin{cases} 1, & X_j = 2, \\ 0, & X_j \neq 2. \end{cases}$$

   Show that $n^{-1} \sum_{j=1}^{n} Y_j$ is an unbiased estimator of p^2. [2 marks]

   (b) Determine the mean squared error of the estimator in part (a). [2 marks]

   (c) It is known that the maximum likelihood estimator of p is $\hat{p} = \frac{1}{2n} \sum_{j=1}^{n} X_j$. Determine the maximum likelihood estimator of p^2. [1 mark]

   (d) Determine the bias of the maximum likelihood estimator of p^2 from part (c). [3 marks]

   (e) For large n, the mean squared error of the maximum likelihood estimator of p^2 is approximately

   $$\frac{2p^3(1-p)}{n}.$$

   Using this approximation, compare the mean squared error of the maximum likelihood estimator with the mean squared error of the estimator in part (a). Which estimator is preferred under this criterion? [2 marks]

Only for STAT2904

5. (10 marks) Let m be known. Suppose A is an unknown non-empty subset of $\{1, \ldots, m\}$. Let $X_1, \ldots, X_n$ be independent random variables with

   $$\mathbb{P}_A(X = x) = \begin{cases} 1/|A|, & x \in A, \\ 0, & x \notin A. \end{cases}$$

   (a) Write down the likelihood function for A. [2 marks]

   (b) Find the maximum likelihood estimator of A. [2 marks]

   (c) Let $K = |A|$, the cardinality of set A. Use the invariance property to find the maximum likelihood estimator of K. [1 mark]

   (d) Is the maximum likelihood estimator of K biased? Justify your answer. [3 marks]

   (e) Is the maximum likelihood estimator of A consistent as $n \to \infty$, with m and A fixed? Justify your answer. [2 marks]